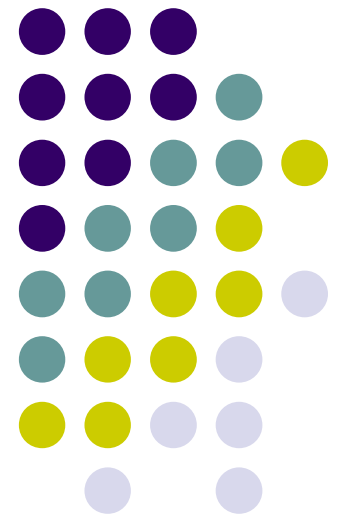
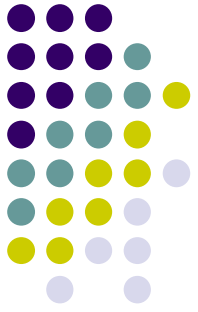


Thiết kế thiết bị đo

Các vấn đề về mạch khách đại



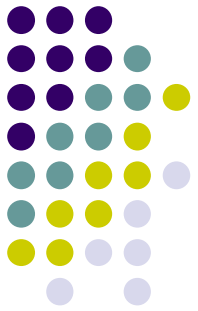
Khuếch đại tín hiệu



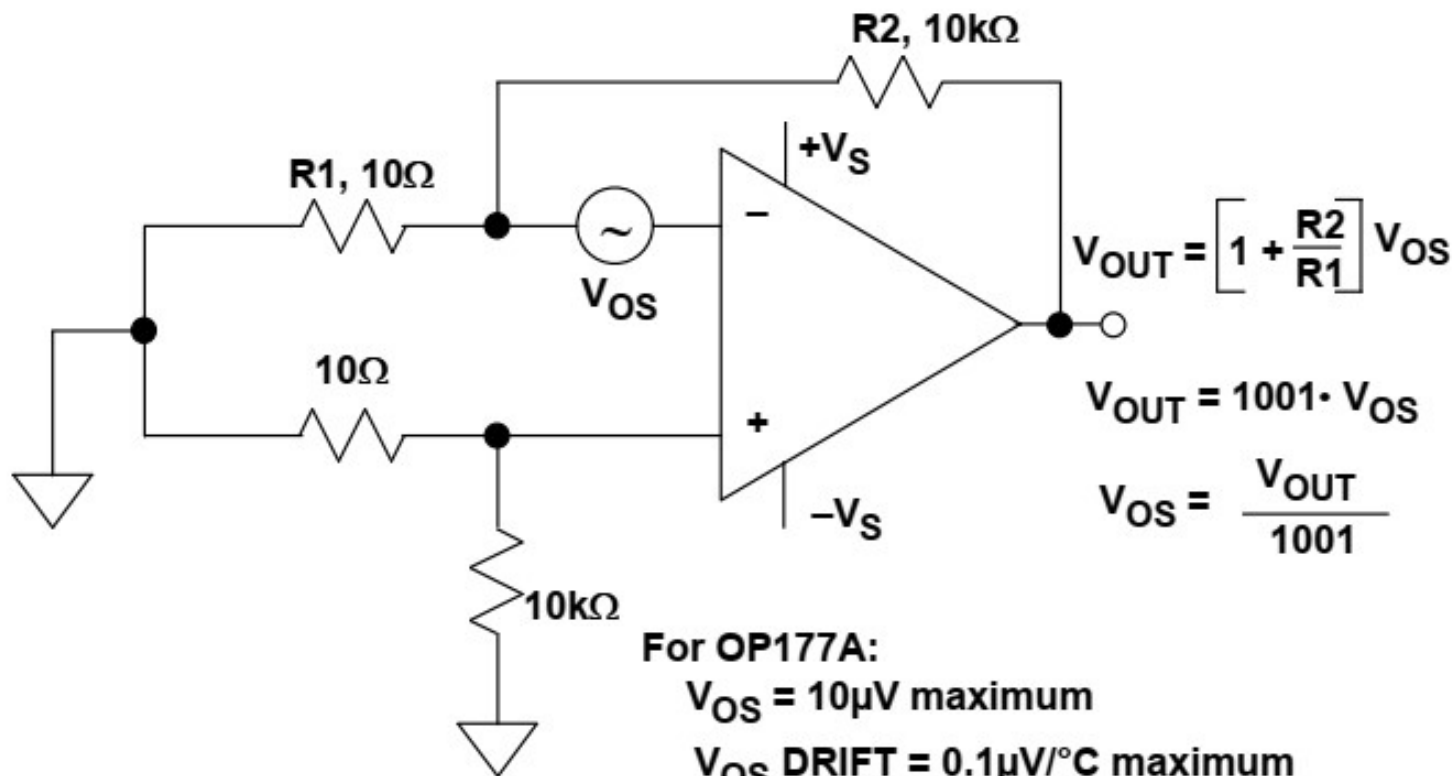
AMPLIFIERS FOR SIGNAL CONDITIONING

- Input Offset Voltage <100 μ V
- Input Offset Voltage Drift <1 μ V/ $^{\circ}$ C
- Input Bias Current <2nA
- Input Offset Current <2nA
- DC Open Loop Gain >1,000,000
- Unity Gain Bandwidth Product, f_u 500kHz - 5MHz
- Always Check Open Loop Gain at Signal Frequency!
- 1/f (0.1Hz to 10Hz) Noise <1 μ V p-p
- Wideband Noise <10nV/ $\sqrt{\text{Hz}}$
- CMR, PSR >100dB
- Single Supply Operation
- Power Dissipation

Khuếch đại tín hiệu



MEASURING INPUT OFFSET VOLTAGE



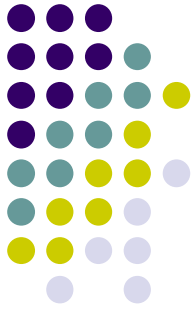
For OP177A:

$V_{OS} = 10\mu V$ maximum

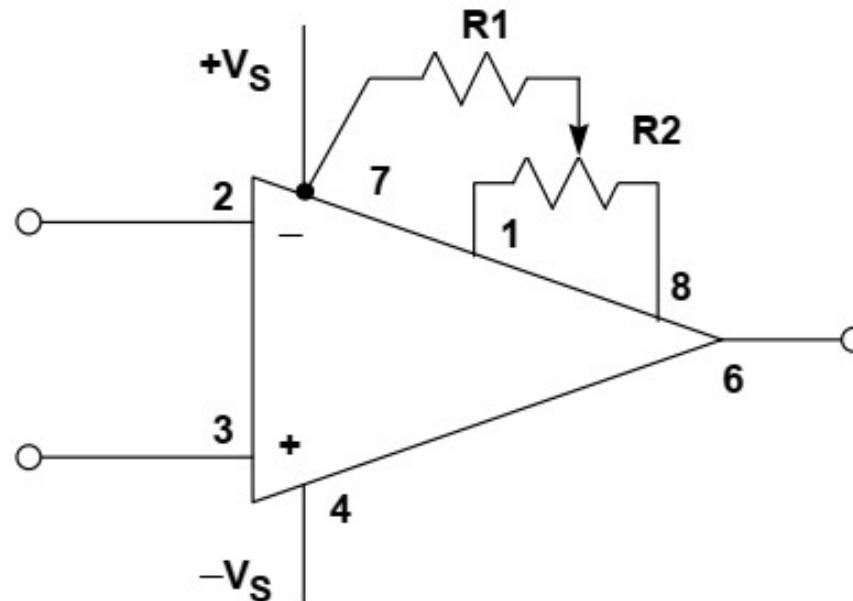
V_{OS} DRIFT = $0.1\mu V/^{\circ}C$ maximum

V_{OS} STABILITY = $0.2\mu V/\text{month}$ typical

Khuếch đại tín hiệu



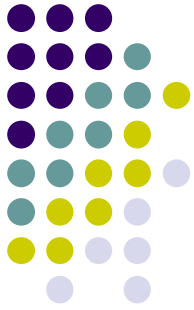
OP177/AD707 OFFSET ADJUSTMENT PINS



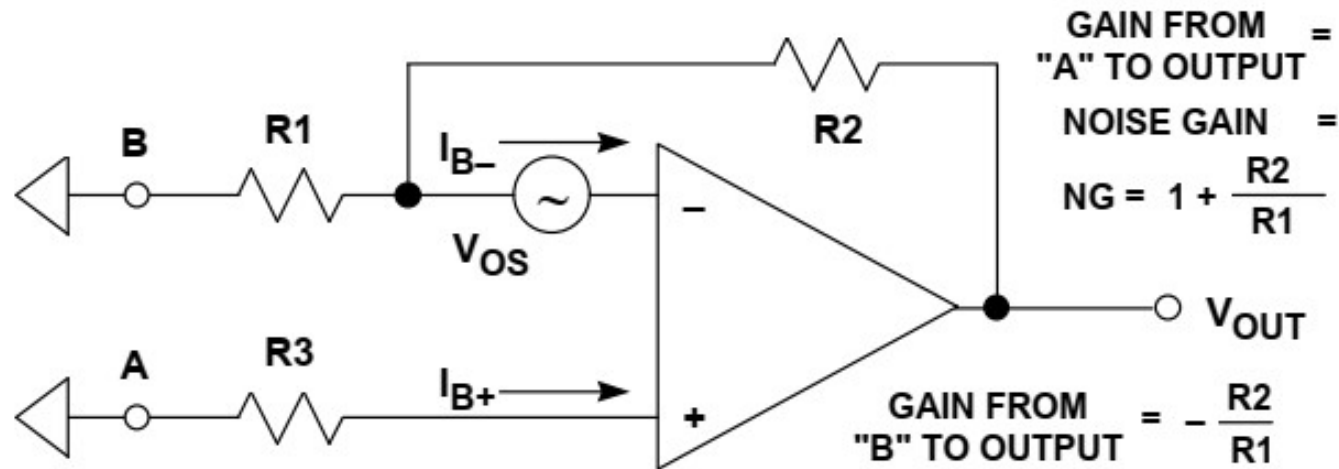
■ $R1 = 10k\Omega$, $R2 = 2k\Omega$, OFFSET ADJUST RANGE = $200\mu V$

■ $R1 = 0$, $R1 = 20k\Omega$, OFFSET ADJUST RANGE = $3mV$

Khuếch đại tín hiệu



OP AMP TOTAL OFFSET VOLTAGE MODEL



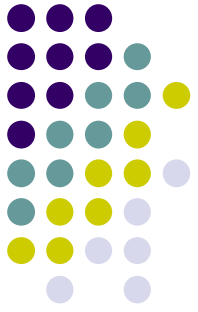
■ $OFFSET (RTO) = V_{OS} \left[1 + \frac{R2}{R1} \right] + I_{B+} \cdot R3 \left[1 + \frac{R2}{R1} \right] - I_{B-} \cdot R2$

■ $OFFSET (RTI) = V_{OS} + I_{B+} \cdot R3 - I_{B-} \left[\frac{R1 \cdot R2}{R1 + R2} \right]$

FOR BIAS CURRENT CANCELLATION:

$OFFSET (RTI) = V_{OS}$ IF $I_{B+} = I_{B-}$ AND $R3 = \frac{R1 \cdot R2}{R1 + R2}$

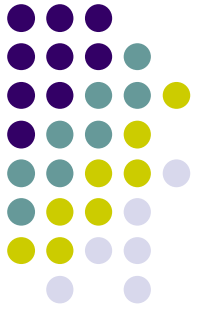
Khuếch đại tín hiệu



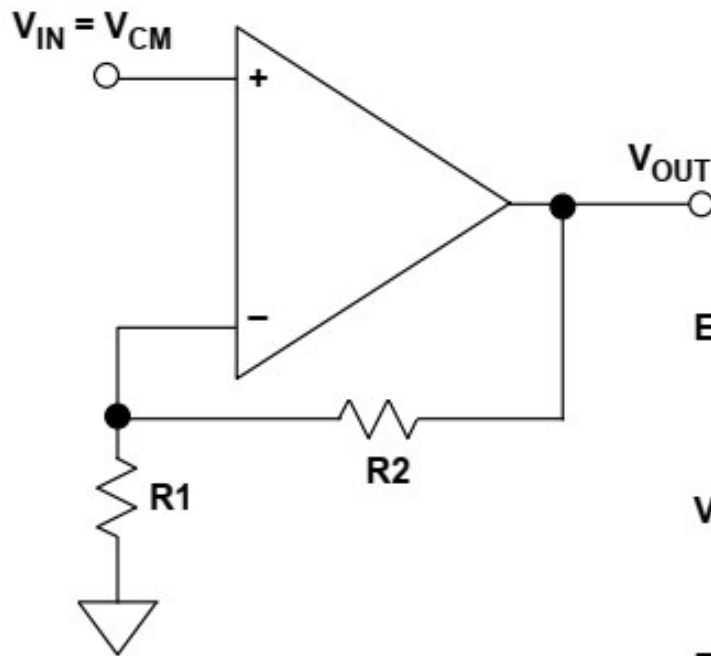
CHANGES IN DC OPEN LOOP GAIN CAUSE CLOSED LOOP GAIN UNCERTAINTY

- "IDEAL" CLOSED LOOP GAIN = NOISE GAIN = NG
- ACTUAL CLOSED LOOP GAIN = $\frac{NG}{1 + \frac{NG}{A_{VOL}}}$
- % CLOSED LOOP GAIN ERROR = $\frac{NG}{NG + A_{VOL}} \times 100\% \approx \frac{NG}{A_{VOL}} \times 100\%$
 - Assume $A_{VOL} = 2,000,000$, $NG = 1,000$
%GAIN ERROR $\approx 0.05\%$
 - Assume A_{VOL} Drops to 300,000
%GAIN ERROR $\approx 0.33\%$
 - CLOSED LOOP GAIN UNCERTAINTY
 $= 0.33\% - 0.05\% = 0.28\%$

Khuếch đại tín hiệu



CALCULATING OFFSET ERROR DUE TO COMMON MODE REJECTION RATIO (CMRR)

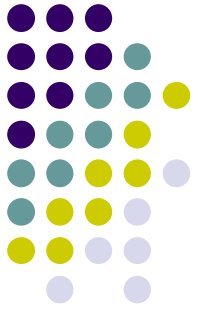


$$\text{ERROR (RTI)} = \frac{V_{CM}}{\text{CMRR}} = \frac{V_{IN}}{\text{CMRR}}$$

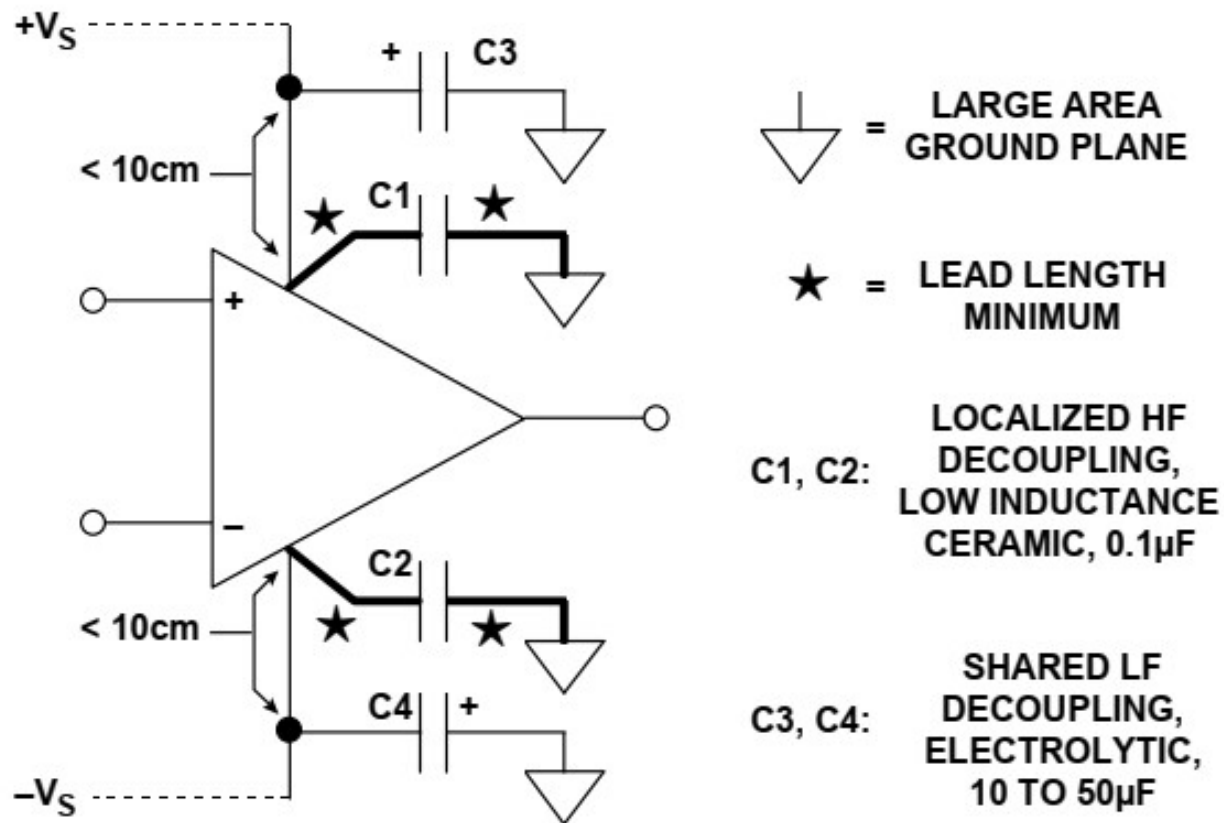
$$V_{OUT} = \left[1 + \frac{R2}{R1} \right] \left[V_{IN} + \frac{V_{IN}}{\text{CMRR}} \right]$$

$$\text{ERROR (RTO)} = \left[1 + \frac{R2}{R1} \right] \left[\frac{V_{IN}}{\text{CMRR}} \right]$$

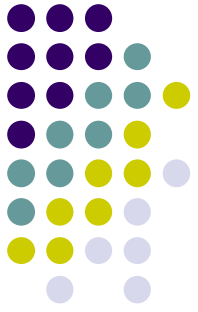
Khuếch đại tín hiệu



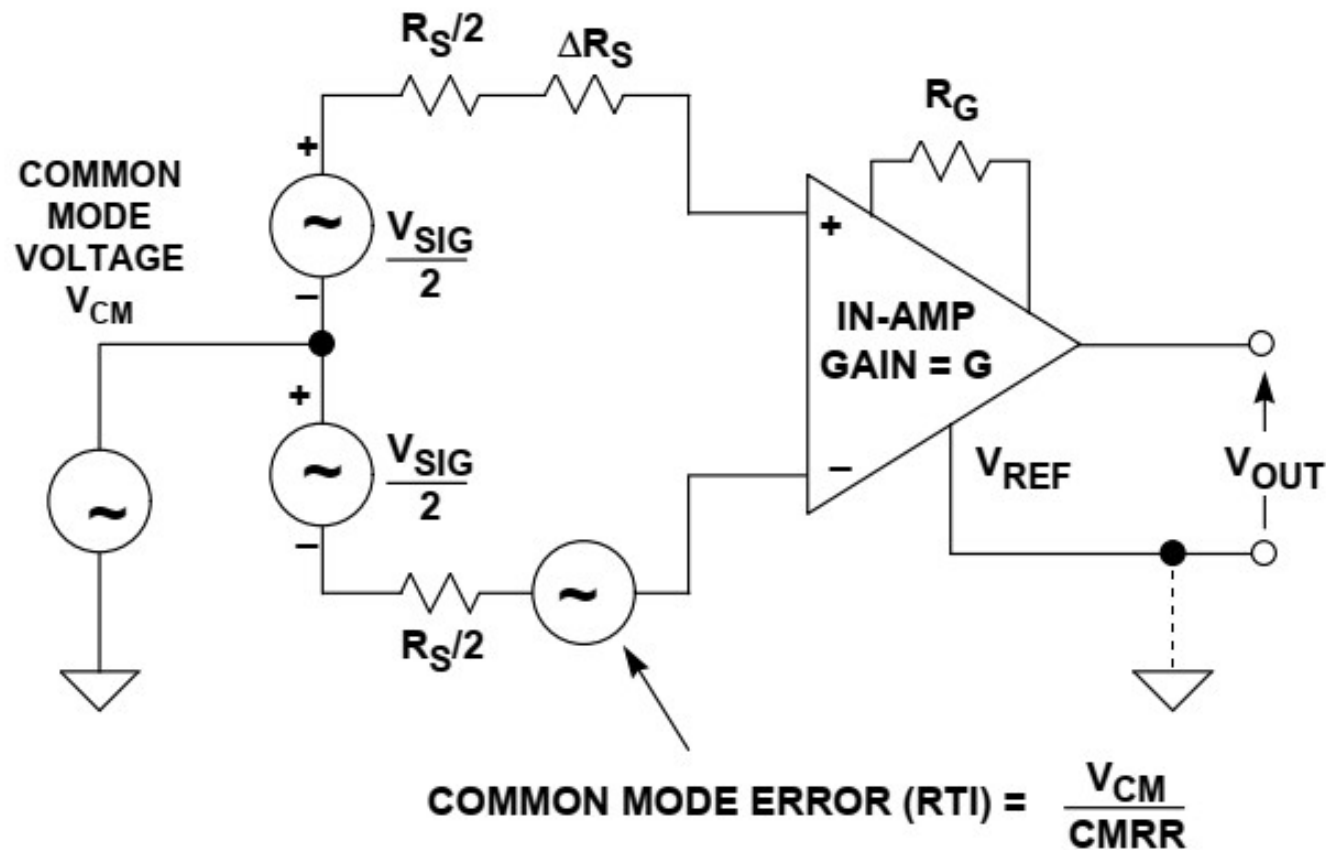
PROPER LOW AND HIGH-FREQUENCY DECOUPLING TECHNIQUES FOR OP AMPS



Khuếch đại tín hiệu



INSTRUMENTATION AMPLIFIER





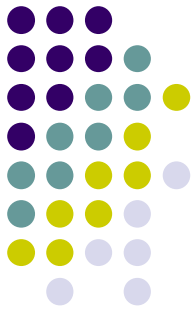
$$G = 1 + \frac{R_2}{R_1} + \frac{2R_2}{R_G}$$

$$\blacksquare V_{OUT} = (V_2 - V_1) \left[1 + \frac{R_2}{R_1} + \frac{2R_2}{R_G} \right] + V_{REF}$$

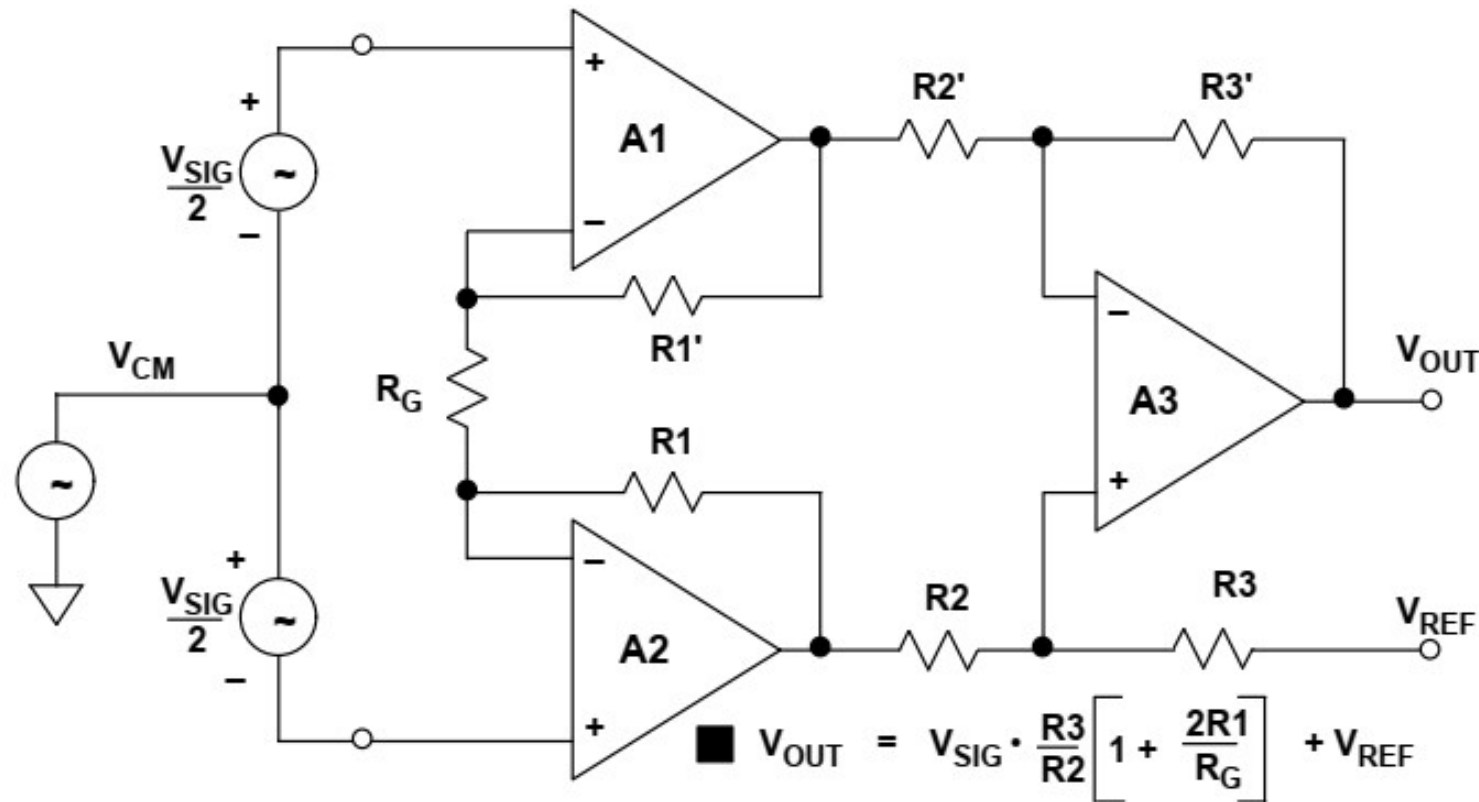
$$\blacksquare V_{OUT} = (V_2 - V_1) \left[1 + \frac{R_2}{R_1} + \frac{2R_2}{R_G} \right] + V_{REF}$$

$$\blacksquare \text{ CMR} \leq 20 \log \left[\frac{\text{GAIN} \times 100}{\% \text{ MISMATCH}} \right]$$

Khuếch đại tín hiệu



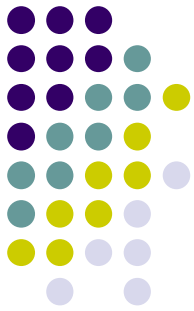
THREE OP AMP INSTRUMENTATION AMPLIFIER



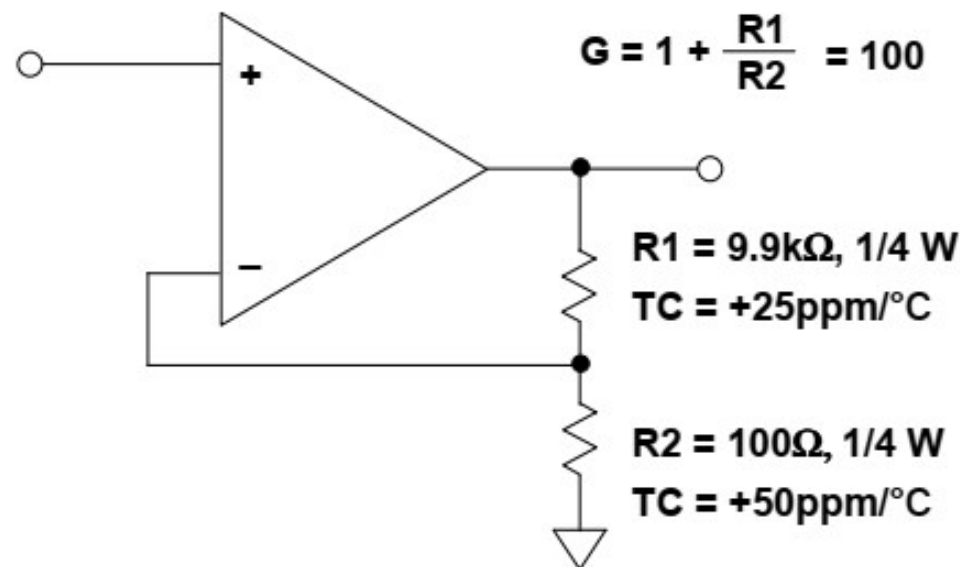
$$\blacksquare CMR \leq 20 \log \left[\frac{GAIN \times 100}{\% \text{ MISMATCH}} \right]$$

$$\blacksquare \text{ IF } R2 = R3, G = 1 + \frac{2R1}{R_G}$$

Khuếch đại tín hiệu

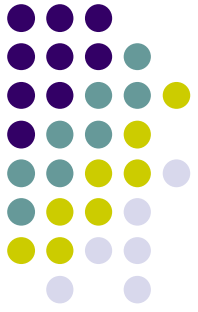


RESISTOR TEMPERATURE COEFFICIENT MISMATCHES CAUSE GAIN VARIATION WITH TEMPERATURE

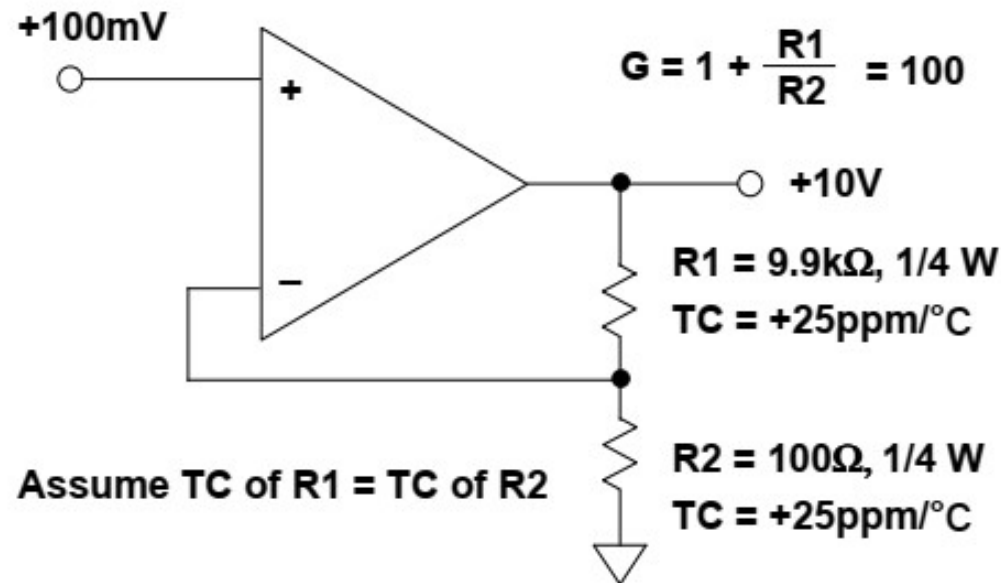


- Temperature change of 10°C causes gain change of 250ppm
- This is 1LSB in a 12-bit system and a disaster in a 16-bit system

Khuếch đại tín hiệu

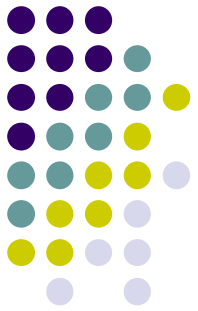


**RESISTOR SELF-HEATING EVEN IN MATCHED RESISTORS
CAN CAUSE GAIN VARIATION WITH INPUT LEVEL**

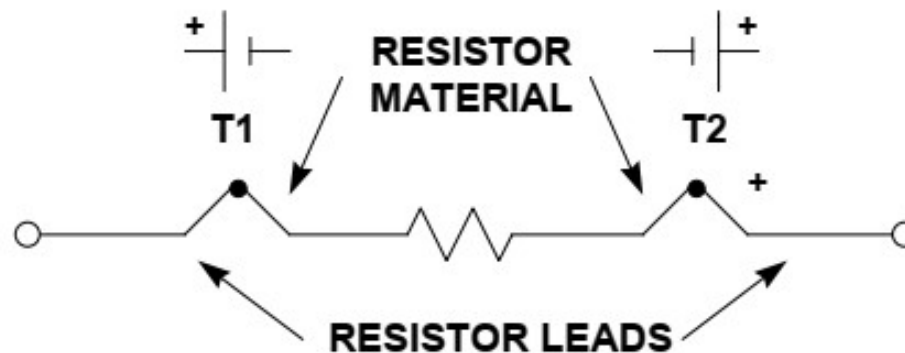


- R1, R2 Thermal Resistance = $125^{\circ}C / W$
- Temperature of R1 will rise by $1.24^{\circ}C$, $P_D = 9.9mW$
- Temperature rise of R2 is negligible, $P_D = 0.1mW$
- Gain is altered by 31ppm, or 1/2 LSB @ 14-bits

Khuếch đại tín hiệu



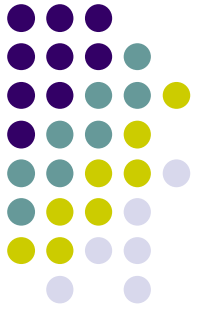
RESISTORS CONTAIN THERMOCOUPLES



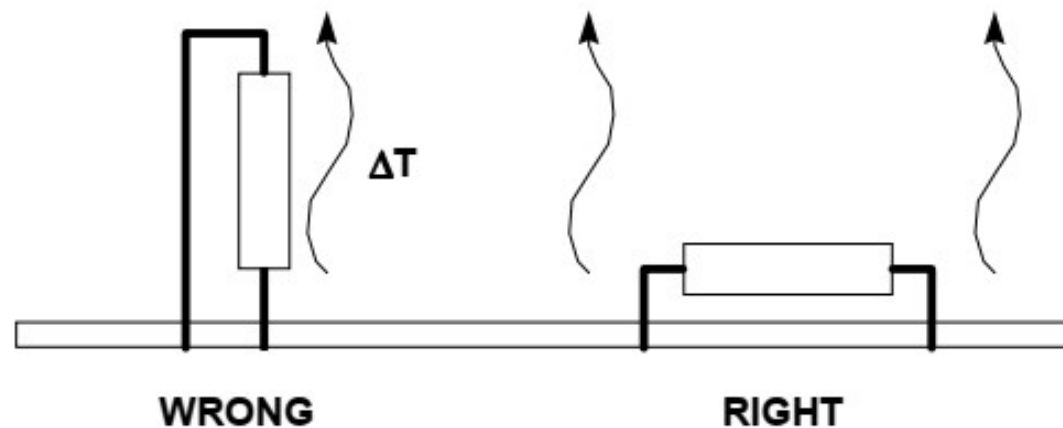
TYPICAL RESISTOR THERMOCOUPLE EMFs

■ CARBON COMPOSITION	$\approx 400 \mu\text{V}/^\circ\text{C}$
■ METAL FILM	$\approx 20 \mu\text{V}/^\circ\text{C}$
■ EVENOHM OR MANGANIN WIREWOUND	$\approx 2 \mu\text{V}/^\circ\text{C}$
■ RCD Components HP-Series	$\approx 0.05 \mu\text{V}/^\circ\text{C}$

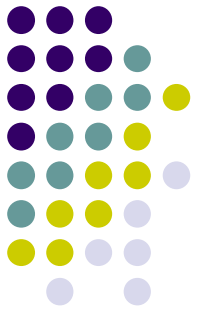
Khuếch đại tín hiệu



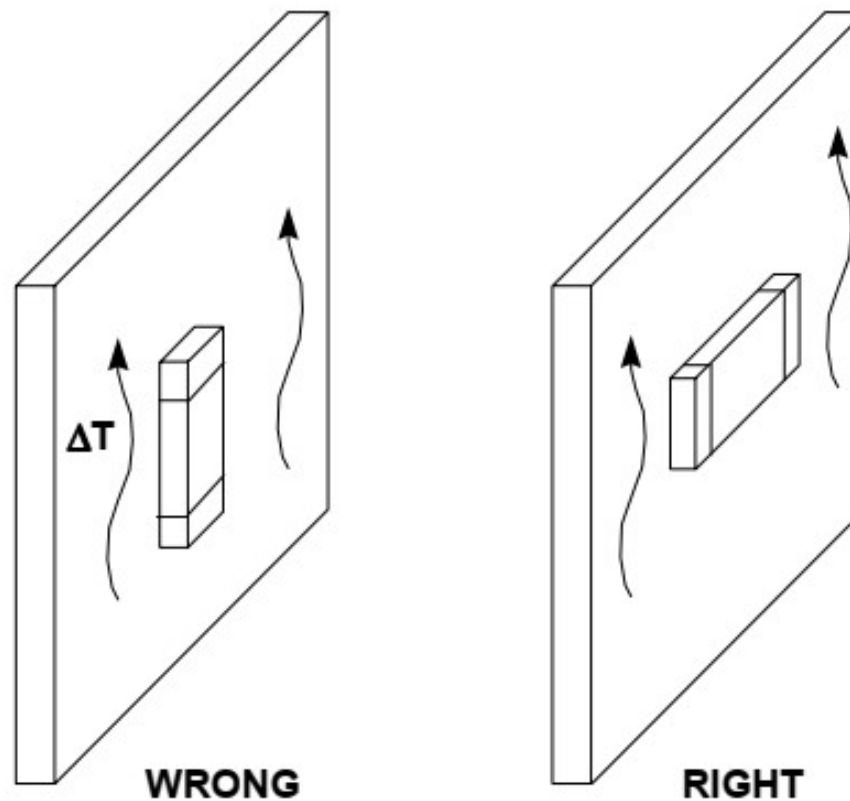
**AVOIDING THERMAL GRADIENTS MINIMIZES
THERMOCOUPLE ERROR VOLTAGES**



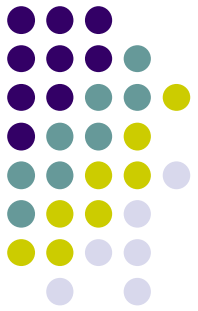
Khuếch đại tín hiệu



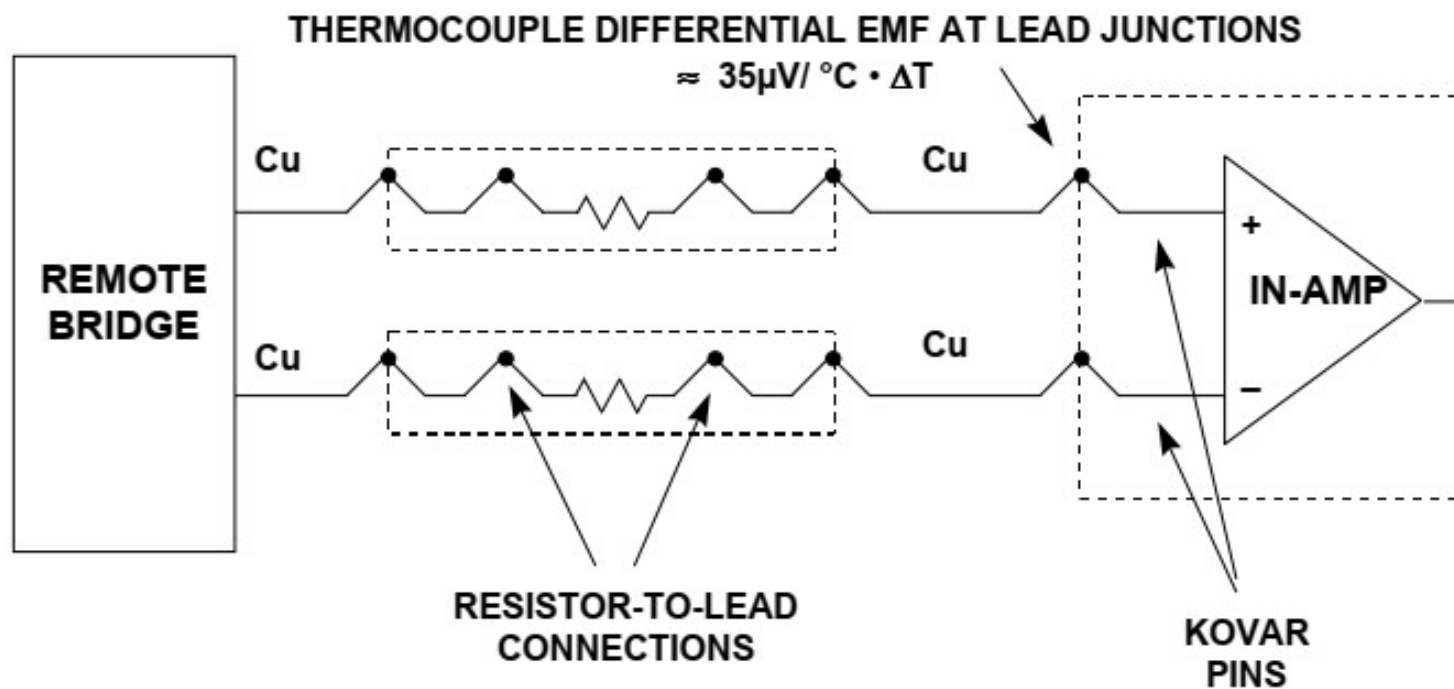
**PROPER ORIENTATION OF SURFACE MOUNT RESISTORS
MINIMIZES THERMOCOUPLE ERROR VOLTAGE**



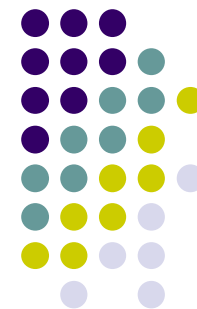
Khuếch đại tín hiệu



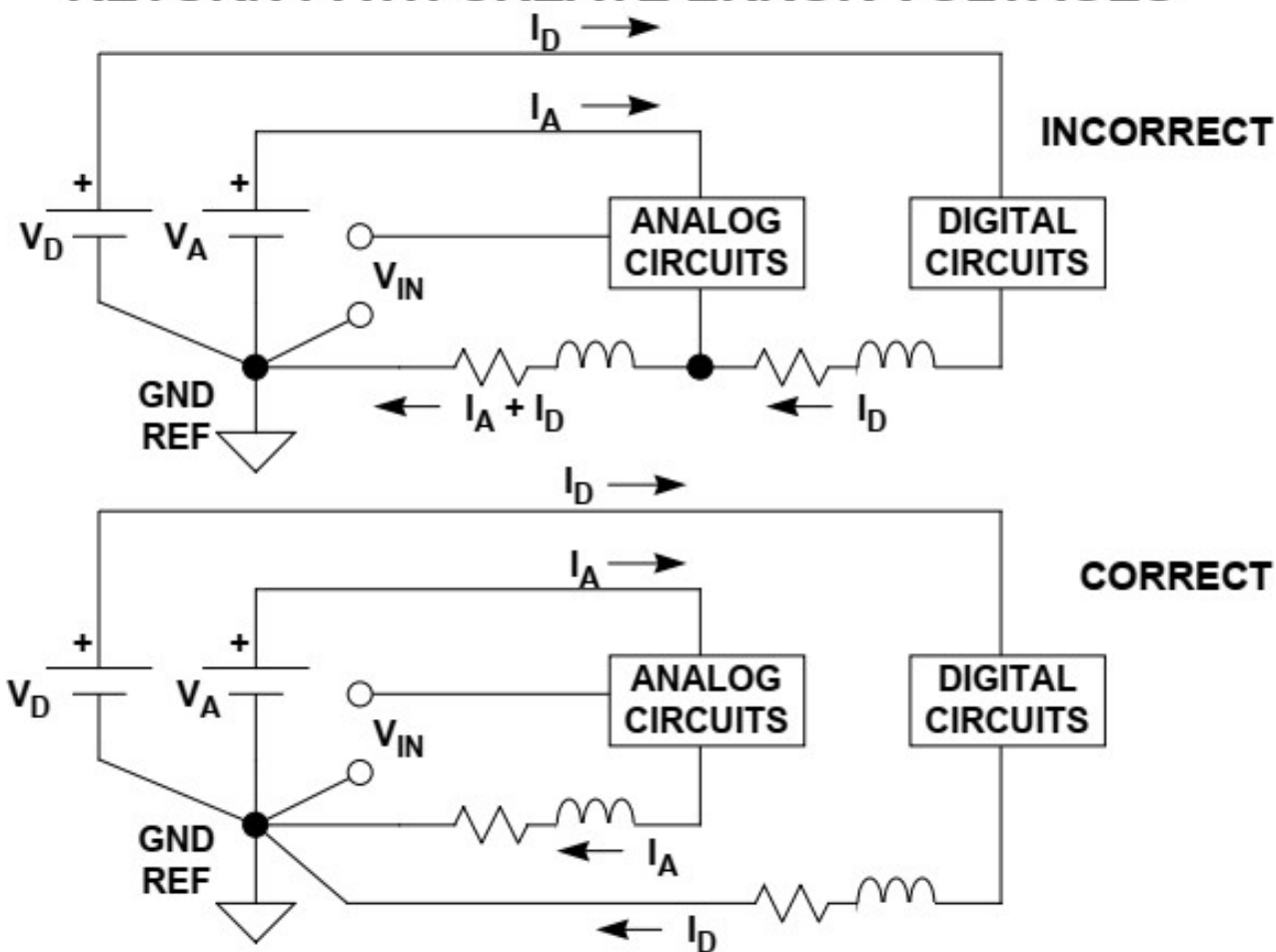
PARASITIC THERMOCOUPLES IN SIMPLE CIRCUIT



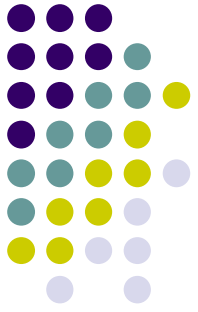
GND tín hiệu



DIGITAL CURRENTS FLOWING IN ANALOG RETURN PATH CREATE ERROR VOLTAGES



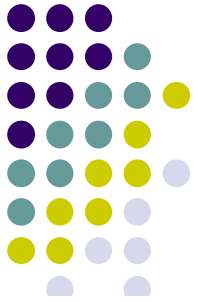
GND tín hiệu



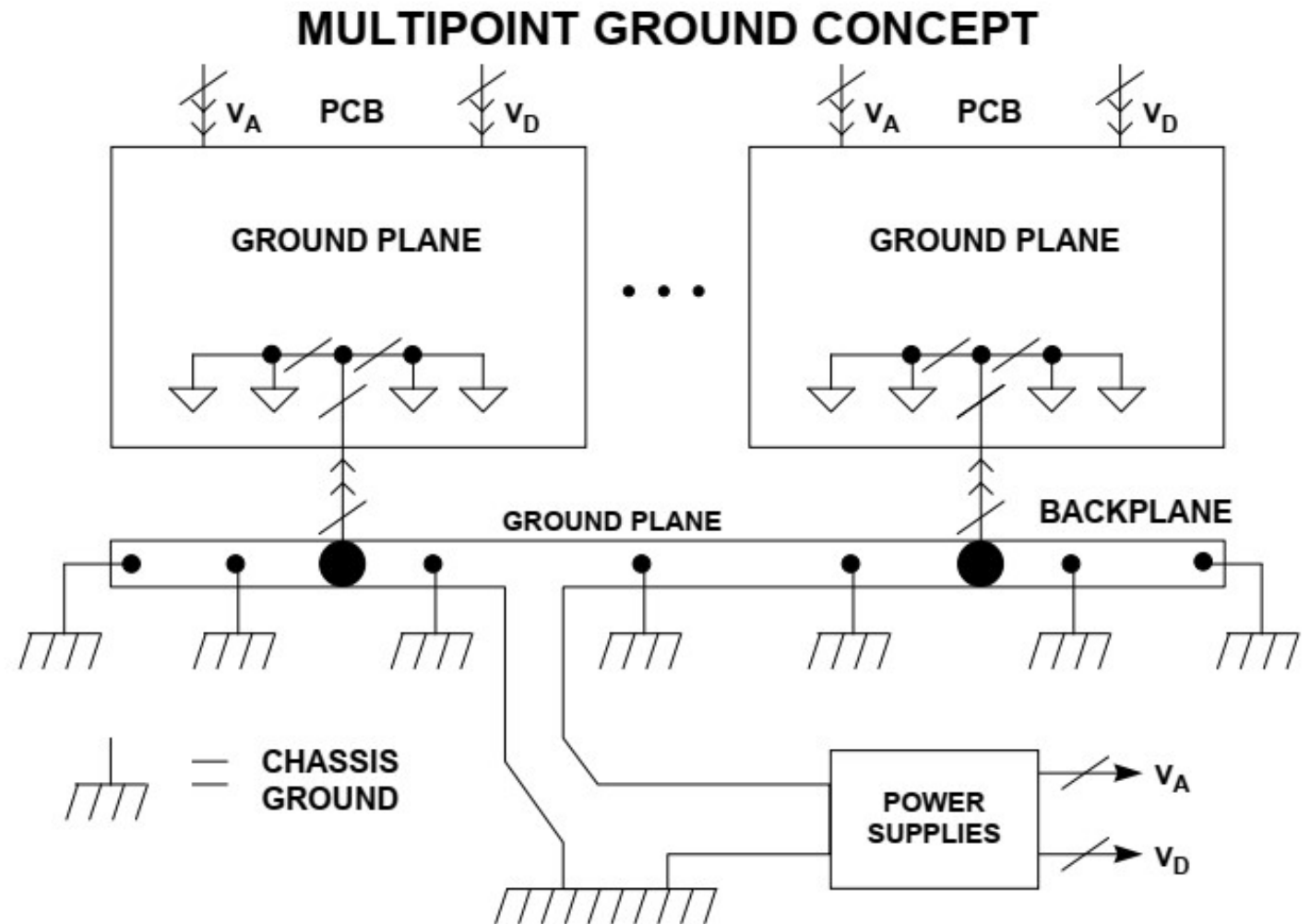
GROUND PLANES ARE MANDATORY!

- **Use Large Area Ground (and Power) Planes for Low Impedance Current Return Paths (Must Use at Least a Double-Sided Board!)**
- **Double-Sided Boards:**
 - ◆ **Avoid High-Density Interconnection Crossovers and Feedthroughs Which Reduce Ground Plane Area**
 - ◆ **Keep > 75% Board Area on One Side for Ground Plane**
- **Multilayer Boards**
 - ◆ **Dedicate at Least One Layer for the Ground Plane**
 - ◆ **Dedicate at Least One Layer for the Power Plane**
- **Use at Least 30% to 40% of PCB Connector Pins for Ground**
- **Continue the Ground Plane on the Backplane Motherboard to Power Supply Return**

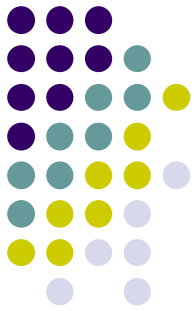
GND tín hiệu



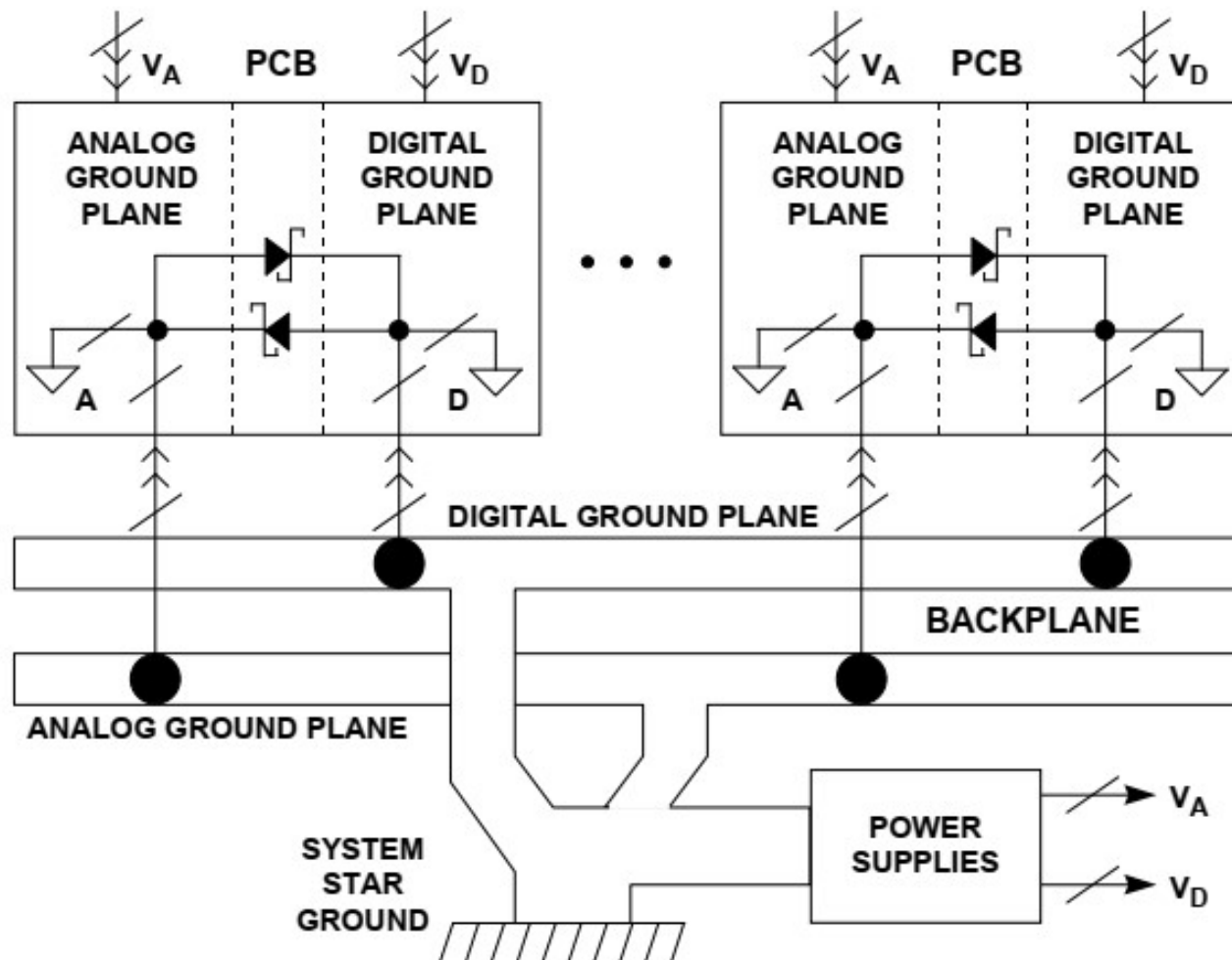
- Multi card



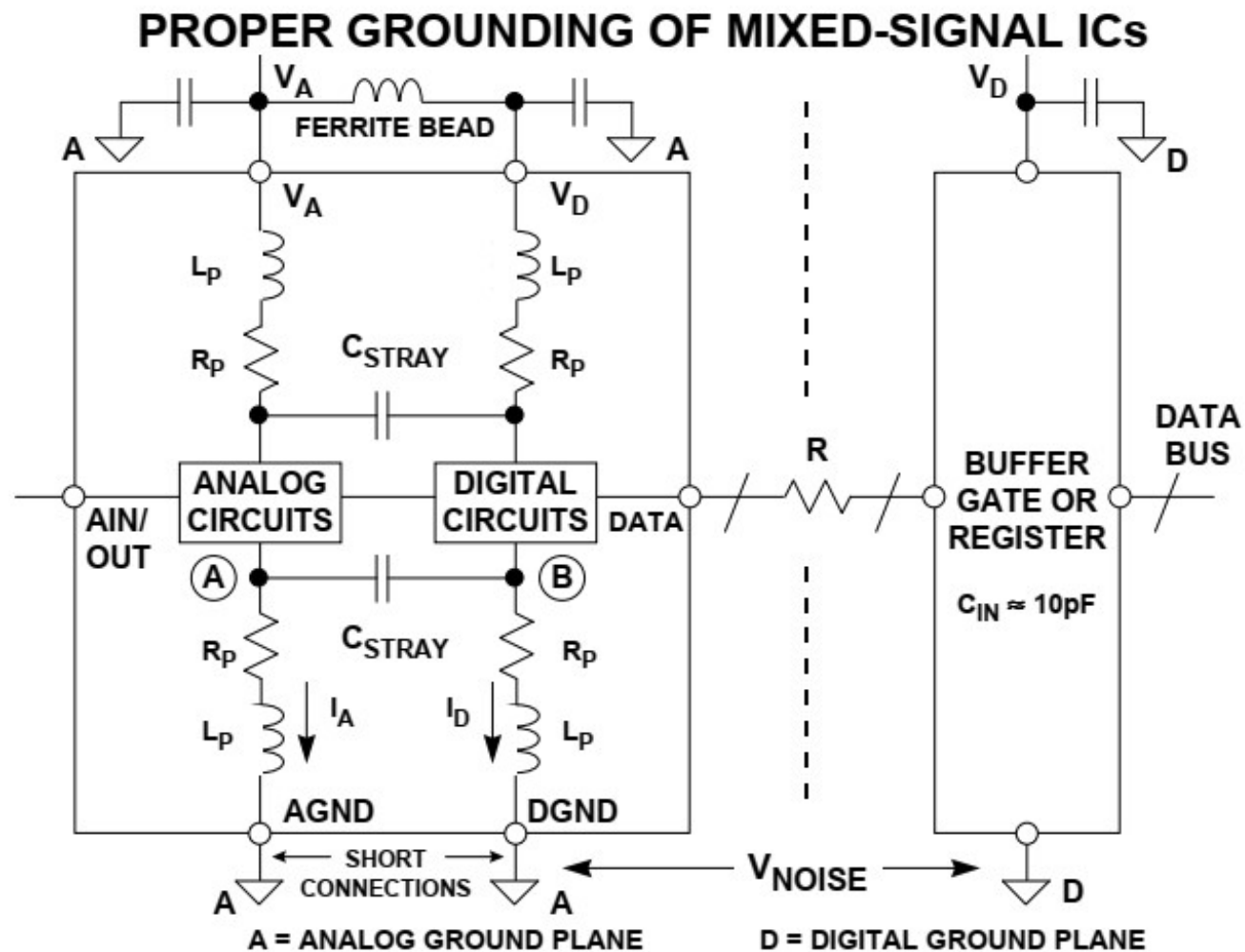
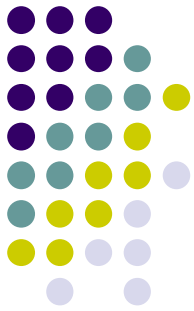
GND tín hiệu



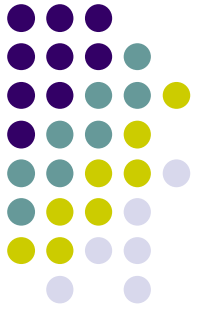
SEPARATING ANALOG AND DIGITAL GROUND PLANES



GND tín hiệu



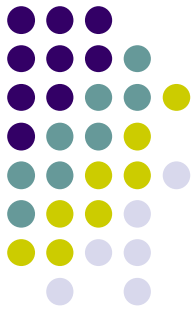
Giảm ồn nguồn và lọc



SWITCHING REGULATOR NOISE REDUCTION TOOLS

- Capacitors
- Inductors
- Ferrites
- Resistors
- Linear Post Regulation
- Proper Layout and Grounding Techniques
- **PHYSICAL SEPARATION FROM SENSITIVE ANALOG CIRCUITS!!**

Giảm ồn nguồn và lọc

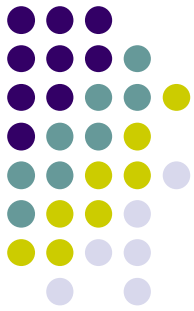


TYPES OF CAPACITORS

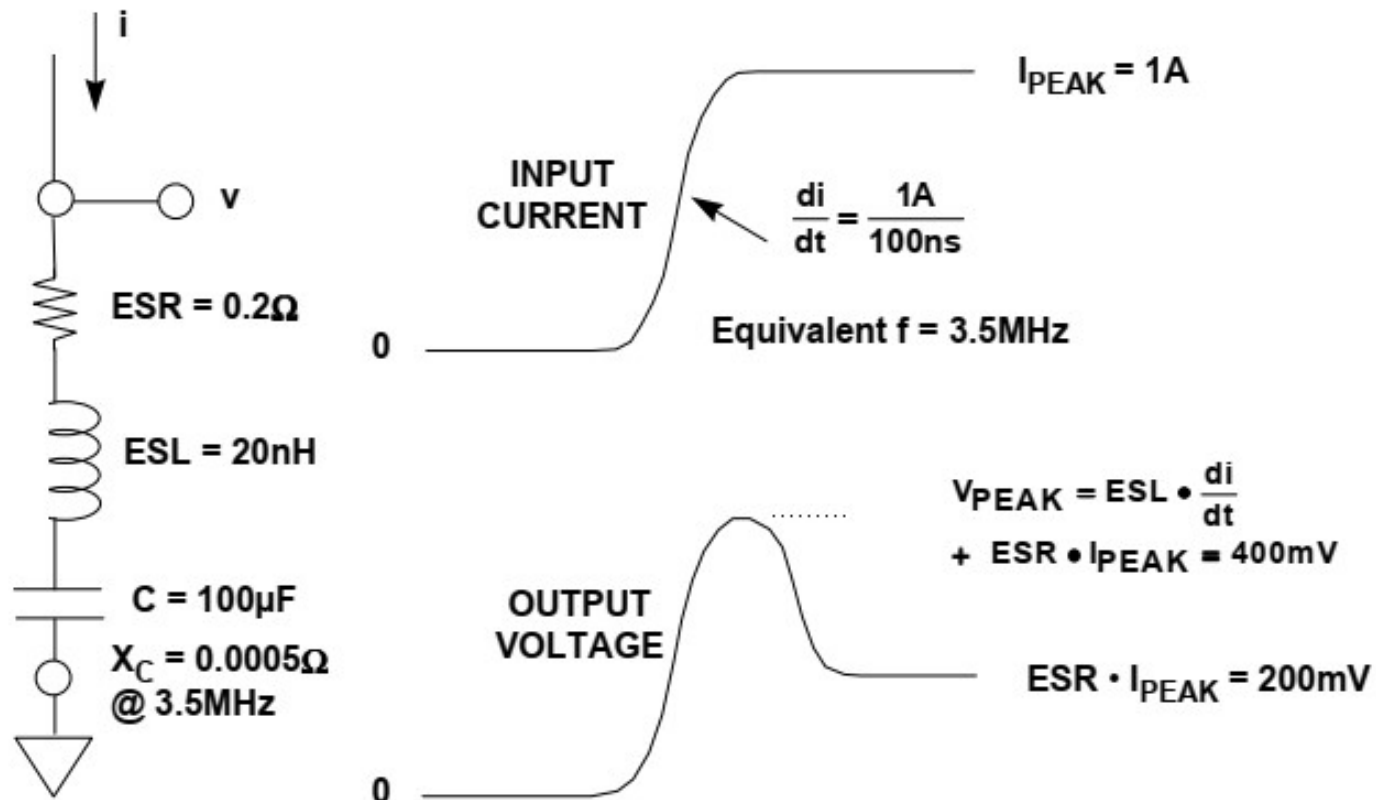
	Aluminum Electrolytic (General Purpose)	Aluminum Electrolytic (Switching Type)	Tantalum Electrolytic	OS-CON Electrolytic	Polyester (Stacked Film)	Ceramic (Multilayer)
Size	100 μF	120 μF	120 μF	100 μF	1 μF	0.1 μF
Rated Voltage	25 V	25 V	20 V	20 V	400 V	50 V
ESR	0.6 Ω @ 100 kHz	0.18 Ω @ 100 kHz	0.12 Ω @ 100 kHz	0.02 Ω @ 100 kHz	0.11 Ω @ 1 MHz	0.12 Ω @ 1 MHz
Operating Frequency (*)	$\cong$ 100 kHz	$\cong$ 500 kHz	$\cong$ 1 MHz	$\cong$ 1 MHz	$\cong$ 10 MHz	$\cong$ 1 GHz

(*) Upper frequency strongly size and package dependent

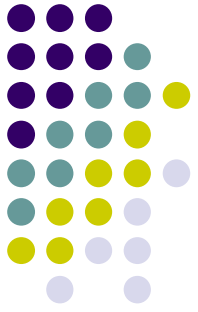
Giảm ồn nguồn và lọc



CAPACITOR EQUIVALENT CIRCUIT AND PULSE RESPONSE



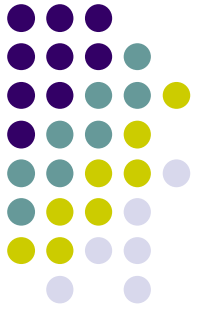
Giảm ồn nguồn và lọc



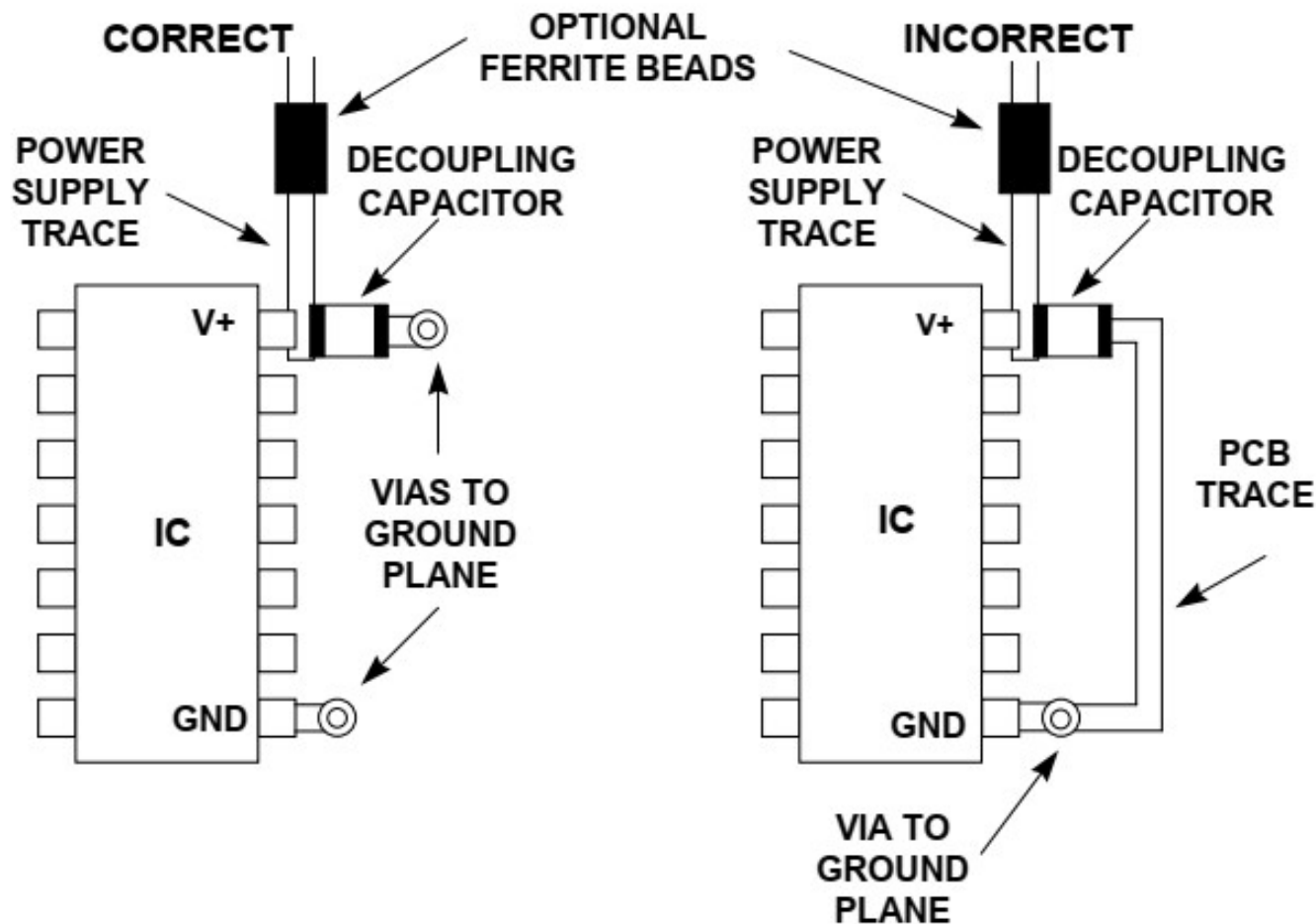
FERRITES SUITABLE FOR HIGH FREQUENCY FILTERS

- Ferrites Good for Frequencies Above 25kHz
- Many Sizes and Shapes Available Including Leaded "Resistor Style"
- Ferrite Impedance at High Frequencies Primarily Resistive -- Ideal for HF Filtering
- Low DC Loss: Resistance of Wire Passing Through Ferrite is Very Low
- High Saturation Current Versions Available
- Choice Depends Upon:
 - ◆ Source and Frequency of Interference
 - ◆ Impedance Required at Interference Frequency
 - ◆ Environmental: Temperature, AC and DC Field Strength, Size / Space Available
- Always Test the Design!

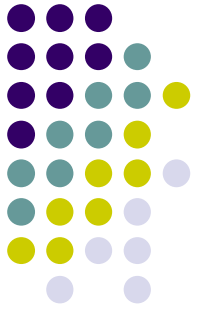
Giảm ồn nguồn và lọc



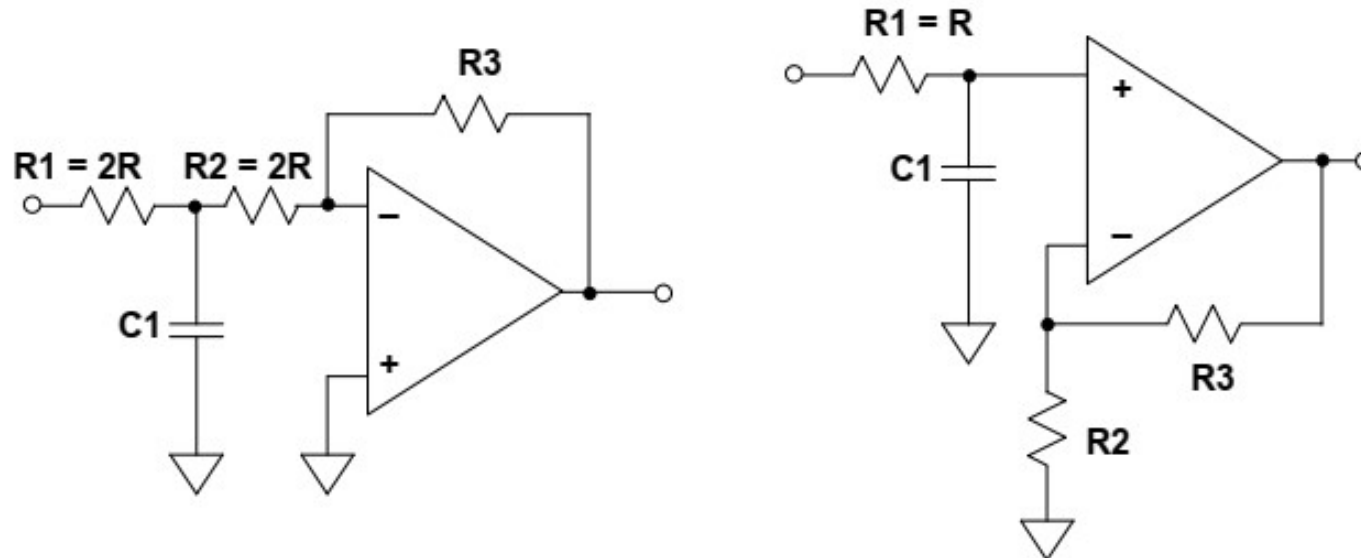
LOCALIZED DECOUPLING TO GROUND PLANE USING SHORTEST PATH



Lọc RFI



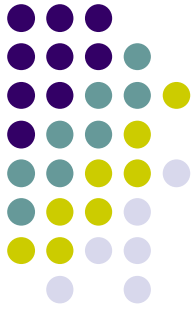
FILTERING AMPLIFIER INPUTS TO PREVENT RFI RECTIFICATION



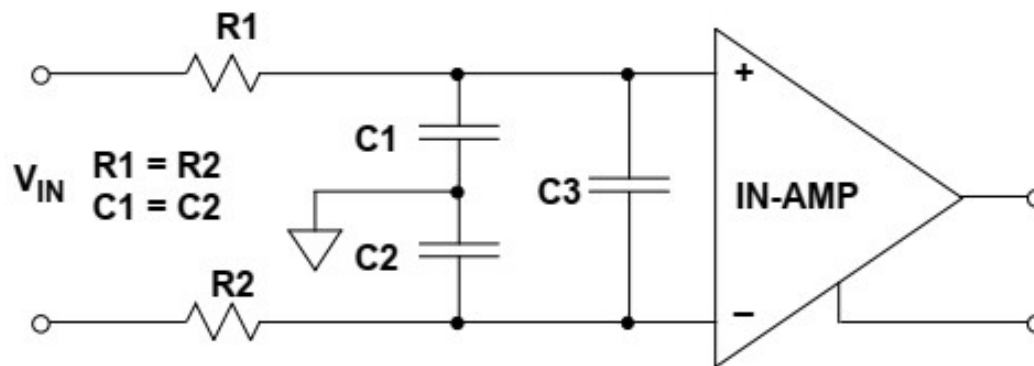
$$\text{FILTER BANDWIDTH} = \frac{1}{2\pi R C1}$$

> 100 × SIGNAL BANDWIDTH

Lọc RFI



FILTERING IN-AMP INPUTS



$$\tau_{\text{DIFF}} = (R1 + R2) C3$$

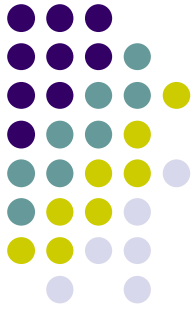
$$\tau_{\text{CM}} = R1 \cdot C1 = R2 \cdot C2 \quad \tau_{\text{DIFF}} \gg \tau_{\text{CM}}$$

R1·C1 SHOULD MATCH R2·C2

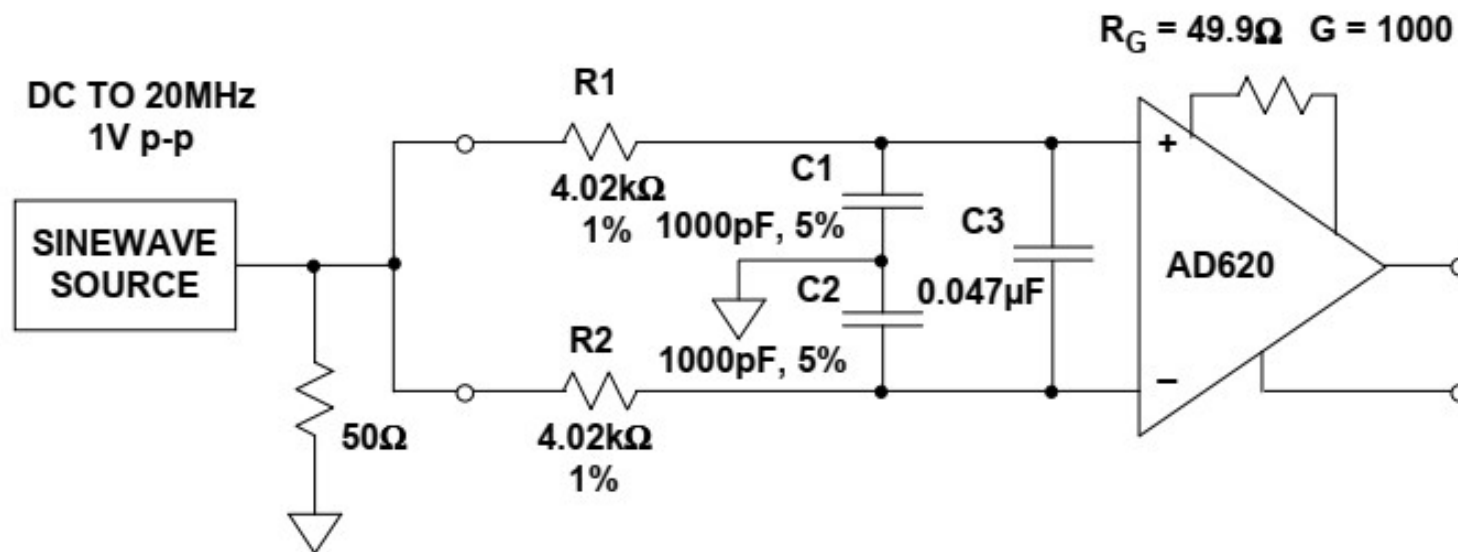
R1 = R2 SHOULD BE 1% RESISTORS
C1 = C2 SHOULD BE 5% CAPACITORS

$$\text{DIFFERENTIAL FILTER BANDWIDTH} = \frac{1}{2\pi (R1 + R2) \left[\frac{C1 \cdot C2}{C1 + C2} + C3 \right]}$$

Lọc RFI

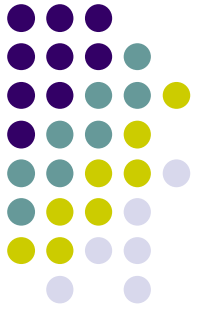


COMMON AND DIFFERENTIAL MODE FILTER WITH AD620

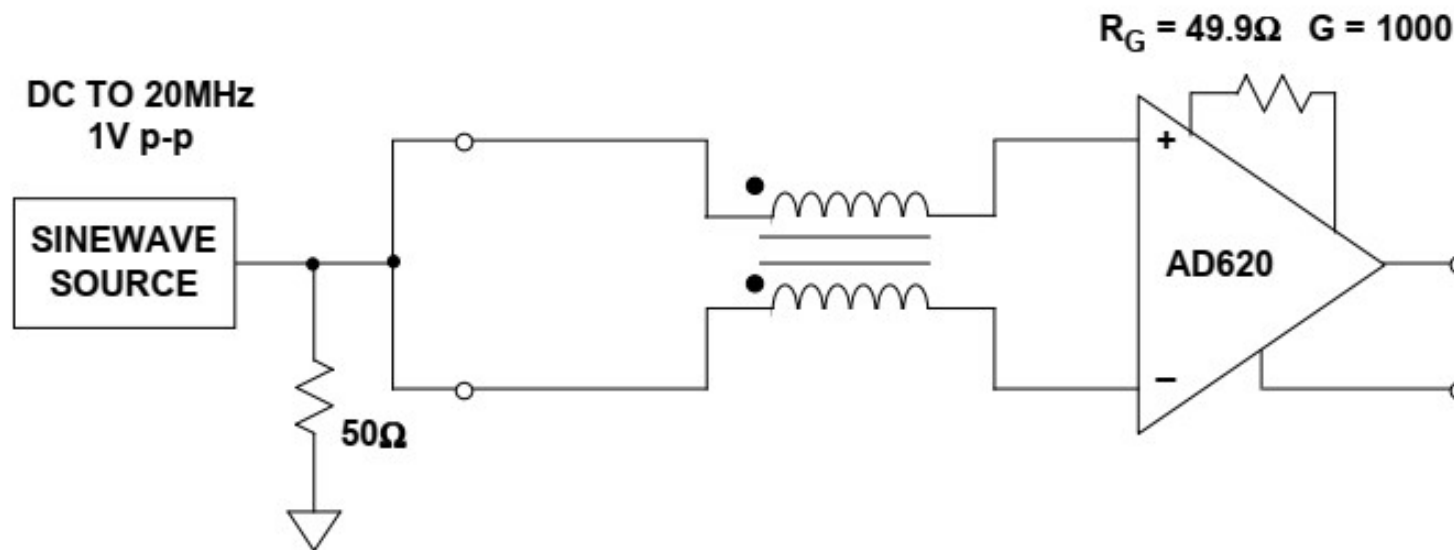


- FILTER BANDWIDTH $\approx 400\text{Hz}$
- OFFSET SHIFT RTI $< 1.5\text{ }\mu\text{V}$

Lọc RFI

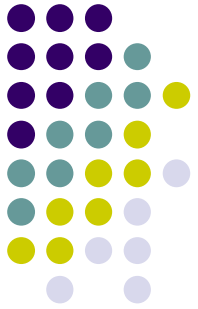


COMMON MODE CHOKE WITH AD620

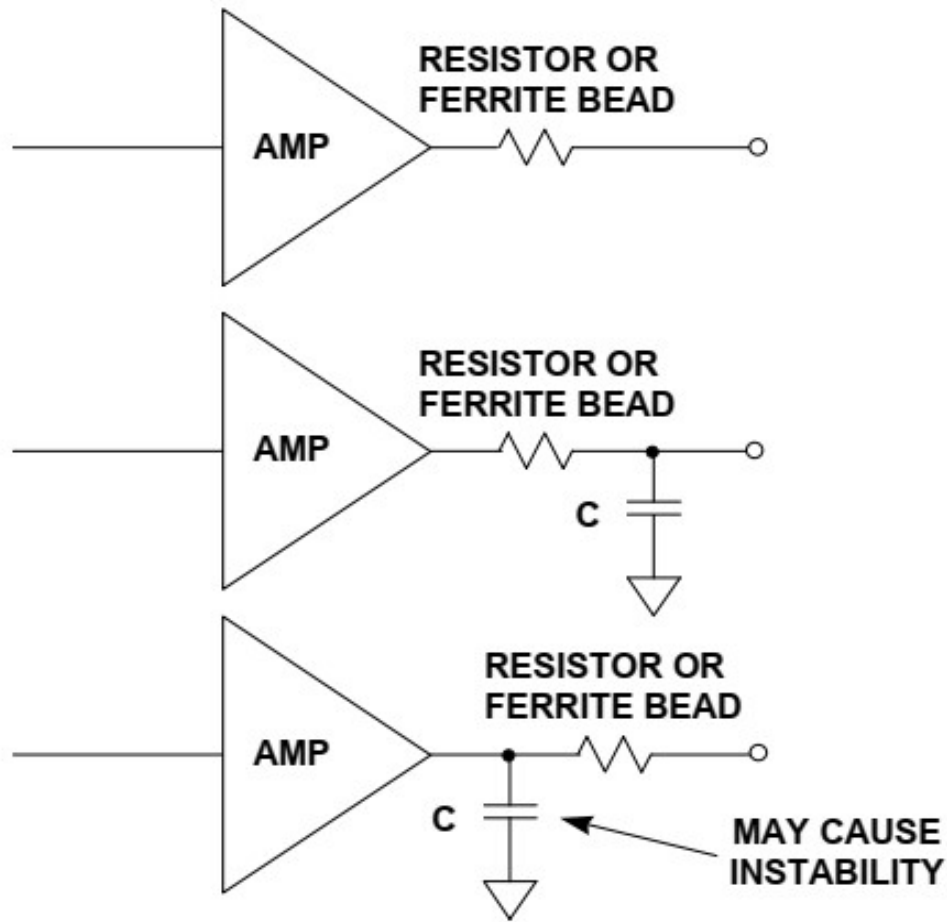


- COMMON MODE CHOKE:
PULSE ENGINEERING B4001
<http://www.pulseeng.com>
- OFFSET SHIFT RTI < 4.5 μV

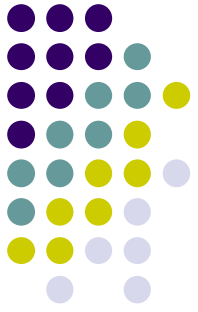
Lọc RFI



FILTERING AMPLIFIER OUTPUTS PROTECTS AGAINST EMI/RFI EMISSION AND SUSCEPTIBILITY



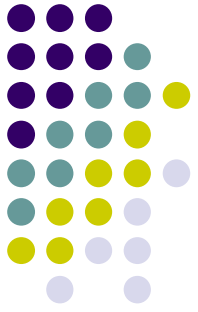
Bảo vệ đầu vào



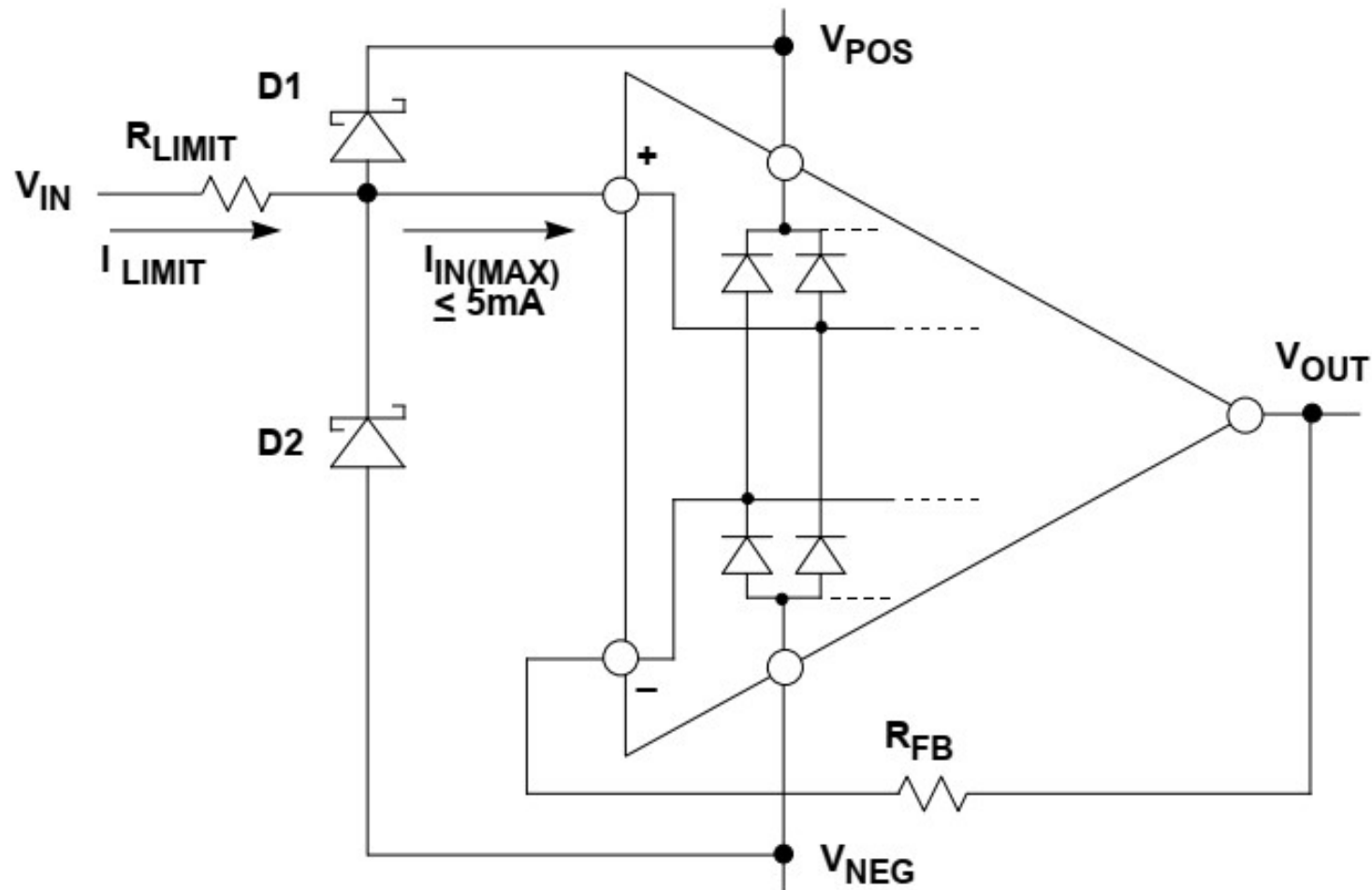
INPUT OVERVOLTAGE

- **INPUT SHOULD NOT EXCEED ABSOLUTE MAXIMUM RATINGS**
(Usually Specified With Respect to Supply Voltages)
- **A Common Specification Requires the Input Signal Remain Within 0.3V of the Supply Rails**
- **Input Stage Conduction Current Should Be Limited**
(Rule of Thumb: $\leq 5\text{mA}$ Unless Otherwise Specified)
- **Avoid Reverse Bias Junction Breakdown in Input Stage Junctions**
- **Differential and Common Mode Ratings May Differ**
- **No Two Amplifiers are Exactly the Same**
- **Some ICs Contain Input Protection (Voltage Clamps, Current Limits, or Both), but Absolute Maximum Ratings Must Still Be Observed**

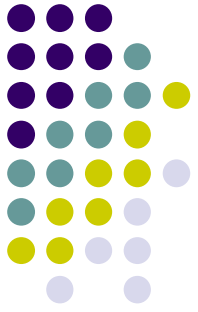
Bảo vệ đầu vào



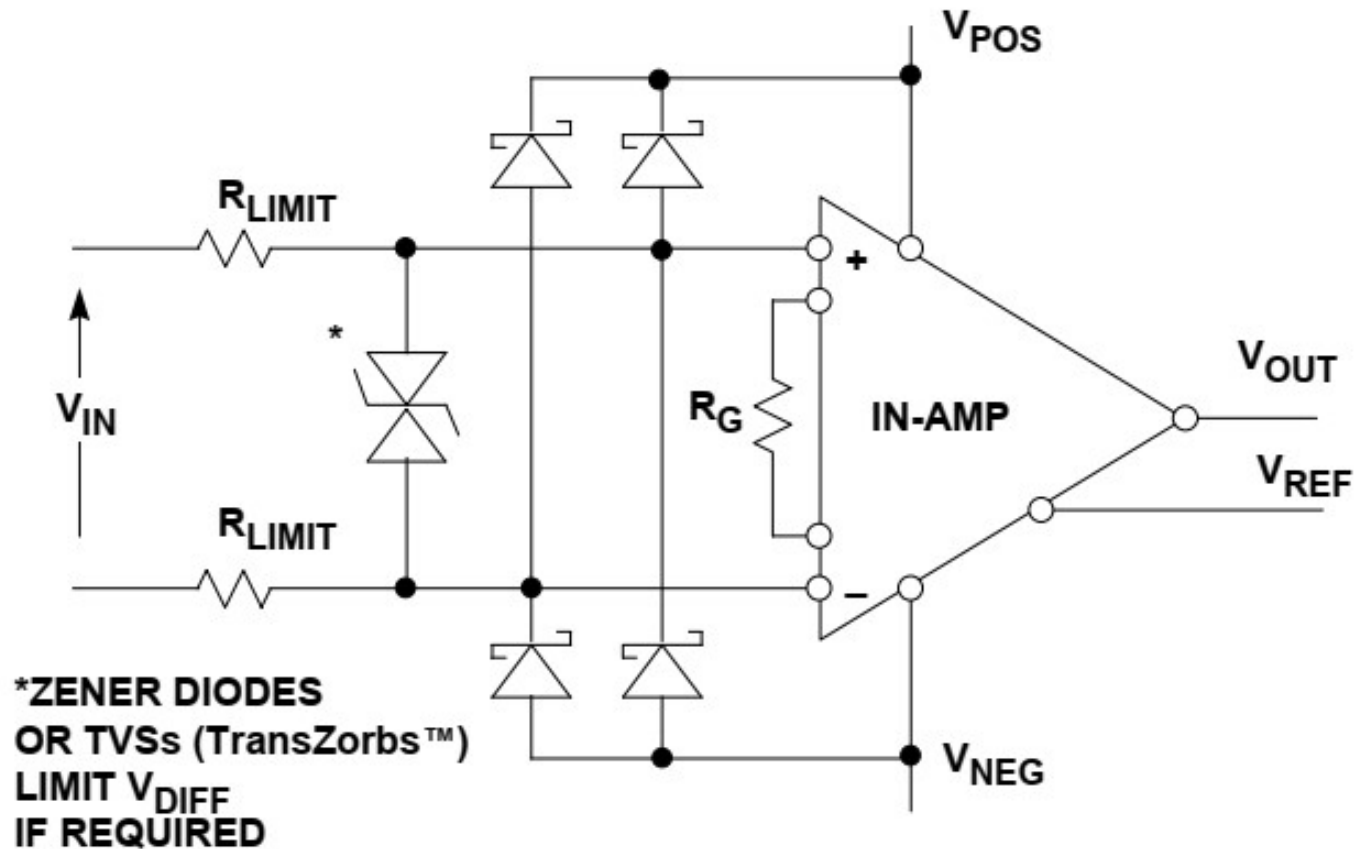
GENERALIZED EXTERNAL OVERVOLTAGE PROTECTION CIRCUIT FOR OP AMPS



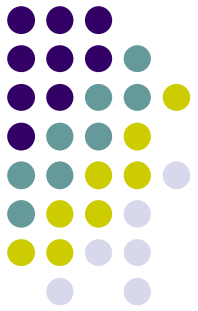
Bảo vệ đầu vào



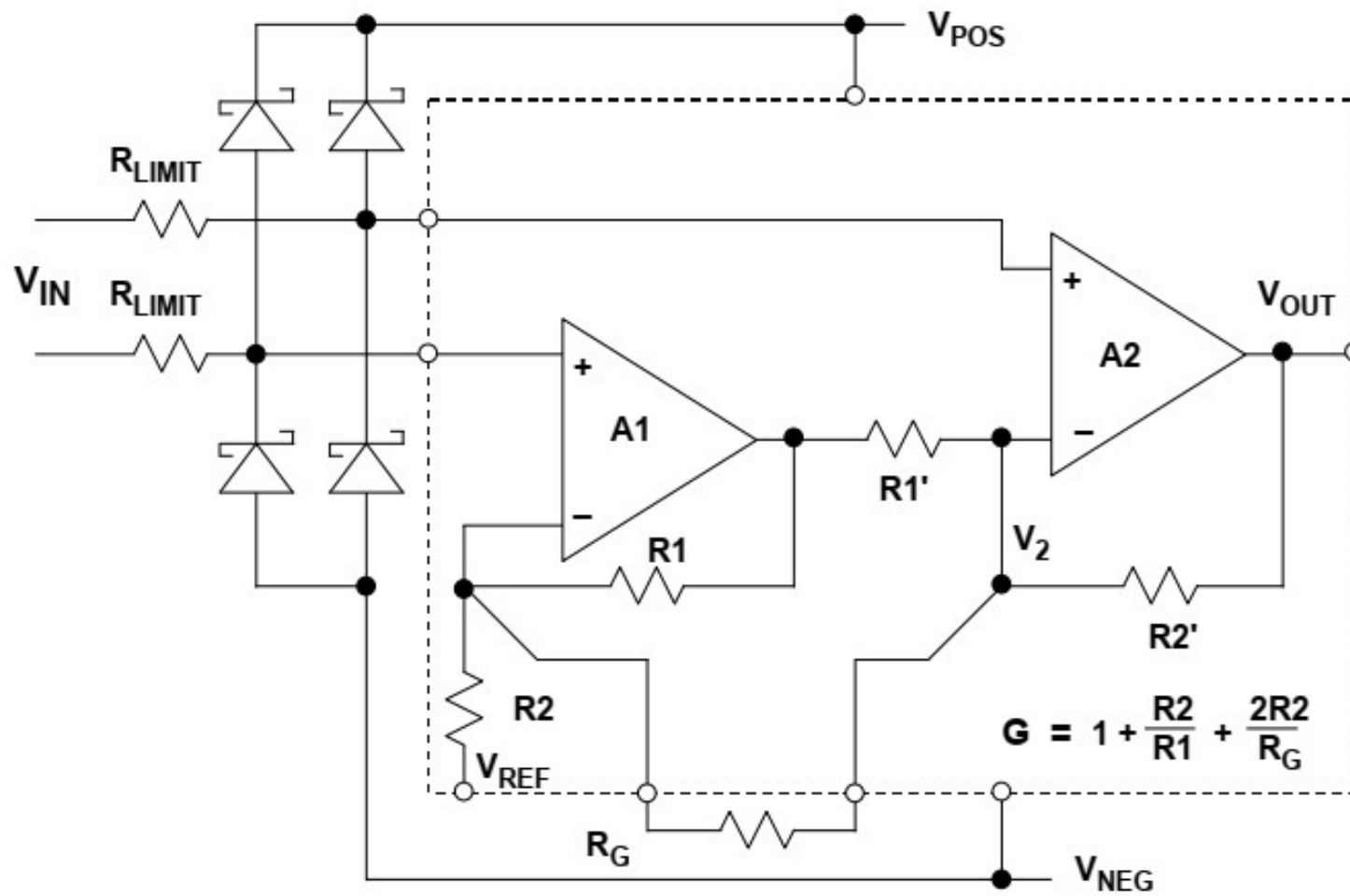
GENERALIZED EXTERNAL PROTECTION FOR INSTRUMENTATION AMPLIFIER INPUTS



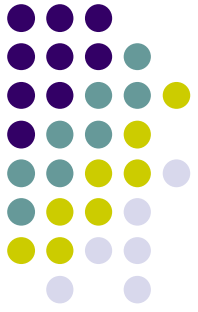
Bảo vệ đầu vào



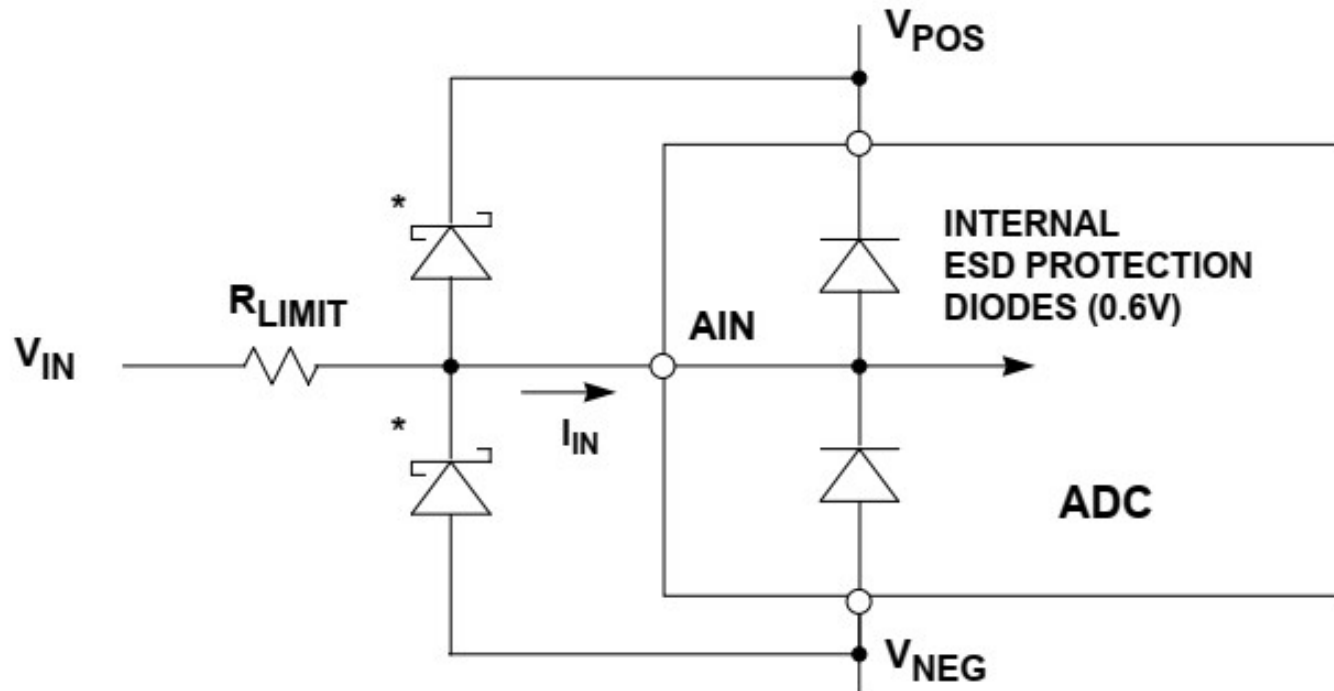
INPUT PROTECTION FOR TWO OP AMP IN-AMP (AD627)



Bảo vệ đầu vào



INPUT PROTECTION FOR ADCs WITH INPUT RANGES WITHIN SUPPLY VOLTAGES



- Choose R_{LIMIT} to Limit I_{IN} Current to 5mA
- *Additional External Schottky Diodes Allow Lower Values of R_{LIMIT}